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FACULTY OF COMPUTER AND INFORMATION SCIENCE

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Noise-Based Wireless Communication Using Software-Defined Radio

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*Write here whom you would like to thank for helping you prepare the thesis.
Be careful not to forget anyone; they might take offense. You can avoid that
by omitting the acknowledgements section entirely.*

To my dear Alenčica.

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List of Abbreviations

kratica	angleško	slovensko
OFDM	Orthogonal Frequency-Division Multiple Access	Ortogonalni frekvenčno deljeni večkratni dostop
MIMO	Multiple-input and multiple-output	večantenski sistem
BER	Bit Error Rate	Stopnja bitnih napak
GRC	GNU Radio Companion	GNU Radio Companion
SNR	Signal-to-Noise Ratio	Razmerje med signalom in šumom
MSNR	Mean Signal-to-Noise Ratio	Varnost na fizičnem nivoju
PLS	Physical Layer Security	Varnost na fizičnem nivoju
SDR	Software Defined Radio	Programsko opredeljena radijska oprema

Summary

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Keywords: computer, computer, computer.

Abstract

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Chapter 1

Introduction

Wireless communication has become an indispensable part of everyday life, enabling high-speed data transmission in mobile telecommunications, satellite communications, and industrial systems. Modern communication standards such as Wi-Fi, Bluetooth, and 4G/5G networks rely on deterministic modulation techniques (variations in amplitude, phase, or frequency) combined with advanced channel coding. Despite their remarkable speed and reliability, these systems share a common limitation: their signals have well-defined structures, making them relatively easy to detect, analyze, and intercept using appropriate equipment.

Consequently, **Physical Layer Security** (*PLS*) has become an increasingly important area of research in modern communication systems. Rather than relying solely on cryptographic protocols implemented at higher layers of the communication stack, the objective is to develop communication schemes that are practically undetectable to unauthorized eavesdroppers. One of the most promising approaches is the use of stochastic¹ signals, or noise-based communication, in which information is not encoded through changes in the carrier frequency but rather through the parameters of the

¹A stochastic signal is a signal whose values vary randomly over time and cannot be described by a deterministic mathematical function. Instead, it is characterized by its statistical properties.

probability distribution governing the underlying random process.

This bachelor's thesis presents a communication system based on the α -stable probability distribution. Unlike conventional modulation schemes, this approach is significantly more difficult to detect and decode, even in the absence of additional cryptographic techniques.

The objective of this thesis is to develop and present a communication method based on the α -stable distribution, describe the corresponding encoding and decoding procedures, and analyze the overall system performance. The thesis investigates the feasibility of using the α -stable distribution as an information carrier and proposes an alternative approach to the development of novel wireless communication methods.

1.1 Challenges and Motivation

Developing a communication system based on the α -stable distribution presents several theoretical and practical challenges:

- **Lack of a closed-form expression:** The probability density function of a general α -stable distribution cannot be expressed using elementary functions, except for a few special cases such as the Gaussian, Cauchy, and Lévy distributions. Consequently, the receiver must estimate the signal parameters from the received samples using computationally intensive numerical methods, significantly increasing the overall computational complexity.
- **Challenges in hardware implementation:** Translating these concepts to actual *Software Defined Radio* (SDR) platforms introduces numerous practical challenges commonly encountered in wireless communication systems. These include accurate time and frequency synchronization of non-deterministic signals, bandwidth limitations, and real-time parameter estimation on hardware platforms with limited computational resources.

It is particularly important to emphasize that, to the best of the author's knowledge, **there is currently no open-source implementation** of a communication system based on α -stable noise operating on real SDR hardware or within the GNU Radio framework.

1.2 Impact and Applications

The results presented in this thesis demonstrate the potential of this communication paradigm and the feasibility of its implementation in practical environments where conventional communication systems may not be sufficient. The proposed approach is particularly suitable for:

- **Covert communications:** Data transmission in environments where the very existence of a communication channel must remain undetected (e.g., military applications or the protection of critical infrastructure).
- **Robust industrial sensing:** Information transmission in highly noisy industrial environments characterized by strong impulsive noise, where conventional communication systems often experience significant performance degradation.

Chapter 2

Current State of Mobile Networks

Modern mobile networks are designed around information transmission using deterministic signals whose properties are defined by various modulation techniques. Technologies used in 4G and 5G networks rely on changing the amplitude, phase, or frequency of the carrier signal, with the aim of using the available spectrum as efficiently as possible while ensuring reliable data transmission. To improve performance, advanced methods such as OFDM, MIMO, and adaptive coding are used.

In the design of such networks, the signal is treated as the information carrier, while noise is an unwanted phenomenon that degrades transmission quality. As a result, much of the development effort is directed toward reducing the influence of noise using coding, filtering, channel estimation, and advanced signal-processing methods.

Despite significant progress, existing communication systems still have certain limitations. As the number of users grows and bandwidth demands increase, the search for alternative communication methods becomes an increasingly important research area. In addition to increasing capacity, security and communication protection are becoming increasingly important. Classical modulation techniques have well-known properties, which makes

them relatively easy to detect, analyze, and intercept with suitable equipment.

This opens opportunities for developing new communication methods in which, in addition to reliable data transmission, low detectability of the communication signal itself is important. The use of stochastic signals as information carriers is one of the more effective approaches, since their random and therefore unpredictable nature makes communication harder to detect and thus increases transmission security.

2.1 History of the α -Stable Distribution

Stable distributions are a family of probability distributions that share some nice properties. They were first described by the French mathematician Paul Lévy in 1925 and are therefore sometimes called Lévy alpha-stable distributions. His work showed that certain distributions preserve their shape even when independent random variables are added together. Such distributions are called stable distributions, and one of them is the α -stable distribution, a natural generalization of the normal distribution. Because of properties such as heavy tails and the ability to model asymmetry, this distribution has very good properties for use in finance, telecommunications, physics, and signal processing, where the classical normal distribution often does not satisfy all criteria for describing extreme data.

Chapter 3

α -Stable Distribution

The α -stable distribution is a continuous distribution. This means that there exists an integrable function p_X , called the *probability density*, such that for every $x \in \mathbb{R}$,

$$F_X(x) = \int_{-\infty}^x p_X(t) dt,$$

where F_X is the distribution function of the random variable X .

The probability density of an α -stable distribution cannot, in general, be expressed with elementary functions. Therefore, this distribution is usually given through its characteristic function $\phi_X(t) : \mathbb{R} \rightarrow \mathbb{C}$:

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} p_X(x) dx, \quad t \in \mathbb{R}.$$

It is given by:

$$\phi(t) = \begin{cases} \exp(i\mu t - \gamma|t|^\alpha (1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2})), & \alpha \neq 1, \\ \exp(i\mu t - \gamma|t| (1 + i\beta \frac{2}{\pi} \operatorname{sign}(t) \ln \frac{\pi\alpha}{2})), & \alpha = 1. \end{cases}$$

As mentioned above, the most important property of α -stable distributions is their stability, so it is also useful to define what this stability means.

Definition: Let X_1 and X_2 be independent copies of the random variable X . The distribution of X is stable if for arbitrary constants $a, b > 0$ there

exist constants $c > 0$ and $d \in \mathbb{R}$ such that

$$aX_1 + bX_2 \stackrel{d}{=} cX + d.$$

This means that a linear combination of two independent random variables with the same stable distribution again belongs to the same family of distributions; only its location and scale may change.

Of course, we cannot see this immediately, but we can nicely connect it with the definition of our characteristic function and prove it.

Proof: Let X be a random variable with an α -stable distribution and let X_1 and X_2 be its independent copies. By independence we have

$$\phi_{aX_1+bX_2}(t) = \phi_X(at)\phi_X(bt),$$

since

$$\phi_{aX_1+bX_2}(t) = \mathbb{E}[e^{it(aX_1+bX_2)}] = \mathbb{E}[e^{itaX_1}e^{itbX_2}] = \mathbb{E}[e^{itaX_1}] \mathbb{E}[e^{itbX_2}] = \phi_{X_1}(at)\phi_{X_2}(bt)$$

For an α -stable distribution ($\alpha \neq 1$), the characteristic function is given by

$$\phi_X(t) = \exp\left(i\mu t - \gamma^\alpha |t|^\alpha \left[1 - i\beta \operatorname{sign}(t) \tan\left(\frac{\pi\alpha}{2}\right)\right]\right).$$

Using the form of the characteristic function with the substitutions $t \mapsto at$ and $t \mapsto bt$, together with the independence of X_1 and X_2 , we obtain:

$$\begin{aligned} \phi_{aX_1+bX_2}(t) &= \phi_X(at)\phi_X(bt) \\ &= \exp\left(i\mu(a+b)t - \gamma^\alpha(a^\alpha + b^\alpha)|t|^\alpha \left[1 - i\beta \operatorname{sign}(t) \tan\left(\frac{\pi\alpha}{2}\right)\right]\right). \end{aligned}$$

We want to show that this is equal to:

$$\phi_{cX+d}(t) \exp\left(i(\mu c + d)t - \gamma^\alpha c^\alpha |t|^\alpha \left[1 - i\beta \operatorname{sign}(t) \tan\left(\frac{\pi\alpha}{2}\right)\right]\right).$$

If we define

$$c = (a^\alpha + b^\alpha)^{1/\alpha},$$

then

$$c^\alpha = a^\alpha + b^\alpha,$$

and then

$$d = \mu(a + b - c)$$

.

Therefore, the above expression can be written as the characteristic function of the random variable $cX + d$, where d is an appropriate constant. Since the characteristic function uniquely determines the distribution, it follows that

$$aX_1 + bX_2 \stackrel{d}{=} cX + d.$$

This shows that the class of α -stable distributions is closed under linear combinations of independent identically distributed random variables, which is exactly the property of stability.

3.1 α -Stable Distribution and the Central Limit Theorem

The central limit theorem is one of the most important theorems in probability theory. It describes the behavior of sums of a large number of independent random variables.

Definition: Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with finite expected value

$$\mathbb{E}[X_i] = \mu$$

and finite variance

$$\text{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1),$$

where the symbol $\xrightarrow{d}$ denotes convergence in distribution. This means that the distribution function of the normalized sum approaches the distribution function of the standard normal distribution. An important property of this result is that the initial distribution of the individual random variables does not have to be normal, but it always converges to a normal distribution. This is why this distribution appears so often in nature and in everyday life.

The connection between the central limit theorem and α -stable distributions lies in their stability under addition. The normal distribution is in fact a special case of our distribution, where

$$\alpha = 2.$$

For other values of the parameter

$$0 < \alpha < 2$$

we can obtain stable distributions with infinite variance. Because of their heavy tails, their sums do not converge to the normal distribution, but to another stable distribution. Therefore, α -stable distributions represent a natural generalization of the central limit theorem.

This connection is described by the generalized central limit theorem. It states that a random variable Z is a stable distribution if and only if there exists a sequence of independent and identically distributed random variables

$$X_1, X_2, X_3, \dots$$

and constants

$$a_n > 0, \quad b_n \in \mathbb{R},$$

such that

$$a_n(X_1 + \cdots + X_n) - b_n \xrightarrow{d} Z.$$

So, if the sum of a large number of independent and identically distributed random variables, after appropriate scaling and shifting, converges to a non-degenerate distribution, that distribution must be stable.

For the classical central limit theorem, finite variance is the key condition for obtaining a normal distribution. The generalized central limit theorem expands this condition and also includes distributions with infinite variance. In this case, the limit distribution is an α -stable distribution, where the parameter α determines the scaling rate of the sum:

$$a_n \propto n^{-1/\alpha}.$$

This is the generalized scaling factor in the central limit theorem. In the CLT, the scaling factor is $\frac{1}{\sqrt{n}}$; since the central limit theorem considers the standard normal distribution, where $\alpha = 2$, we see that this is indeed only a generalization.

With this, α -stable distributions are the only possible limit distributions when a large number of independent and identically distributed random variables are summed.

Chapter 4

Using the α -Stable Distribution as an Information Carrier

Now that we have covered the necessary theory, we can understand how it can be used for data transmission. In classical wireless communication, we use deterministic signal properties such as changes in amplitude, phase, or frequency, and thereby transmit our data. In communication based on the α -stable distribution, information is represented by the statistical properties of the signal.

With the α -stable distribution, this works by choosing its parameters at the transmitter and generating a sequence of samples:

$$X_1, X_2, \dots, X_n \sim S_\alpha(\beta, \gamma, \mu)$$

At the receiver, we then try to determine these parameters. Because of the stability property, these parameters can also be estimated approximately, and this property gives us the ability to infer an approximate distribution from multiple random samples, from which the transmitter generated the samples.

Why is a single sample not enough? Due to randomness, a single random variable does not contain enough information to reliably determine an individual parameter. The receiver therefore analyzes multiple samples and estimates the statistical properties of their distribution. In other words, we

estimate the distribution with which parameters were used by the transmitter.

How is this related to the generalized central limit theorem? In a communication channel, the received signal usually represents a combination of many random influences, so it can be written as a sum:

$$Y = X_1 + X_2 + \cdots + X_n.$$

The classical central limit theorem says that such sums converge to a normal distribution, but only when the random variables have finite variance. In phenomena where rare large amplitudes or impulsive noise occur, this assumption is no longer satisfied.

The generalized central limit theorem extends this result and states that normalized sums of independent and identically distributed random variables converge to stable distributions:

$$a_n(X_1 + \cdots + X_n) - b_n \xrightarrow{d} Z.$$

The limit distribution Z is therefore an α -stable distribution, where the parameter α determines the scaling behavior of the sum. For $\alpha = 2$ we obtain the special case of the normal distribution, while for values $\alpha < 2$ we obtain heavy-tailed distributions that make it possible to describe extreme values.

This explains why α -stable distributions are suitable for modeling communication signals. If a signal is composed of a large number of random influences, its statistical form naturally converges toward a stable distribution. Because stability ensures that the family of distributions is preserved under the addition of independent influences, the receiver can still estimate the parameters of the distribution and thus obtain the information encoded in it.

Chapter 5

Simulations

After presenting the mathematical background, we now focus on the implementation of the communication system and the simulation results. The simulations were carried out using GNU Radio Companion (GRC), a graphical development environment for designing, implementing, and testing communication systems.

GNU Radio Companion allows a communication system to be constructed from interconnected processing blocks.

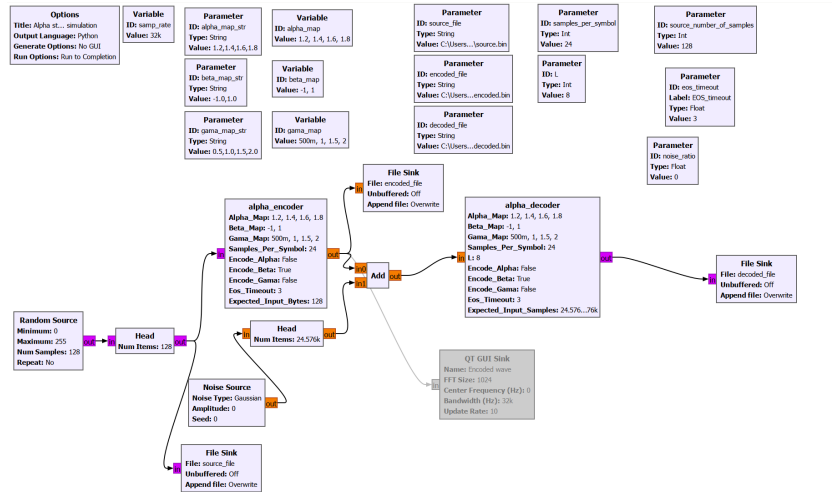


Figure 5.1: GNU Radio flow graph

The flow graph used in this work is shown in Figure 5.1. A random source first generates 128 samples of 8-bit values, resulting in a total of 1024 transmitted bits. The generated bits are stored in an external file before being passed to the encoder, which maps the information into an α -stable signal. The encoded signal is then transmitted through the communication channel, where additive noise can optionally be introduced. At the receiver, the signal is decoded and the recovered bits are stored in a second file. Finally, the transmitted and received files are compared to determine the Bit Error Rate (BER).

Before discussing the simulation results, it is necessary to understand the operation of both the encoder and the decoder.

5.1 Parameters

5.2 Encoder

The encoder receives a stream of input bytes, which are first converted into a sequence of individual bits. These bits are stored in a buffer, from which the required number of bits is read according to the selected modulation configuration to form a single symbol.

The number of bits represented by each symbol depends on the number of possible values assigned to the parameter β . The implementation is also designed to support future extensions in which the parameters α and γ may also carry information. For each parameter, the number of possible values must be a power of two so that every bit combination can be mapped uniquely onto a parameter value. Consequently, the number of bits encoded by a parameter is given by

$$n = \log_2(M),$$

where M denotes the number of entries in the mapping table. The total number of bits represented by one symbol is therefore equal to the sum of

the bits encoded by the individual parameters.

In the implementation presented in this thesis, only the parameter β is used to convey information, while α and γ remain constant. The value of β is selected according to the current combination of input bits.

For every symbol, a predefined number of samples is generated from the corresponding α -stable distribution. This number is controlled by the parameter `samples_per_symbol`. These generated samples constitute the output of the encoder and are transmitted through the communication channel to the next processing block in GNU Radio Companion.

Signal generation is performed using the Chambers–Mallows–Stuck (CMS) algorithm, defined by

$$X = \gamma C \frac{\sin(\alpha V + B)}{(\cos V)^{1/\alpha}} \left(\frac{\cos((1 - \alpha)V - B)}{W} \right)^{\frac{1-\alpha}{\alpha}}, \quad (5.1)$$

where

$$B = \arctan \left(\beta \tan \frac{\pi\alpha}{2} \right), \quad (5.2)$$

and

$$C = \left(1 + \beta^2 \tan^2 \frac{\pi\alpha}{2} \right)^{\frac{1}{2\alpha}}. \quad (5.3)$$

V is a uniformly distributed random variable,

$$V \sim \mathcal{U} \left(-\frac{\pi}{2}, \frac{\pi}{2} \right), \quad (5.4)$$

and W is an exponentially distributed random variable with unit mean,

$$W \sim \text{Exp}(1). \quad (5.5)$$

For each transmitted symbol, the encoder generates `samples_per_symbol` samples according to these equations. The parameters α , β , and γ are selected from the input bits according to the predefined mapping tables.

5.3 Decoder

The decoder receives samples from the α -stable distribution and stores them in a buffer until enough samples have been collected to represent one symbol, i.e., until `samples_per_symbol` samples have been received. Since the encoder always generates a fixed number of samples for each symbol, the decoder can easily separate consecutive symbols.

Each symbol is divided into L equally sized segments. For every segment, the maximum and minimum sample values are determined. These values are then used to compute the mean and variance of the maxima and minima:

$$Y_{\max} = \frac{1}{L} \sum_{l=1}^L Y_{l,\max}, \quad (5.6)$$

$$S_{\max}^2 = \frac{1}{L-1} \sum_{l=1}^L (Y_{l,\max} - Y_{\max})^2, \quad (5.7)$$

$$Y_{\min} = \frac{1}{L} \sum_{l=1}^L Y_{l,\min}, \quad (5.8)$$

$$S_{\min}^2 = \frac{1}{L-1} \sum_{l=1}^L (Y_{l,\min} - Y_{\min})^2. \quad (5.9)$$

These statistics are then used to estimate the parameter β according to

$$\hat{\beta} = 1 - \frac{2}{\exp(\hat{\alpha}(S_{\max} - S_{\min}))}, \quad (5.10)$$

where $\hat{\alpha}$ is estimated as

$$\hat{\alpha} = \frac{\pi}{2\sqrt{6}} \left(\frac{1}{Y_{\max}} + \frac{1}{Y_{\min}} \right). \quad (5.11)$$

The estimated value is compared with the predefined values in the β mapping table, and the closest value is selected. The index of the selected value represents the decoded bit sequence, which is stored in a bit buffer. Once eight bits have been collected, the decoder combines them into a byte and forwards it to the output.

5.4 Simulation Results

The simulations measured the number of incorrectly decoded bits during the transmission of 1024 bits through the proposed communication channel. To reduce the influence of statistical fluctuations, each simulation was repeated 100 times, resulting in a total of 102,400 transmitted bits being analyzed. During the experiments, the values of the parameters β and γ , the noise level, and the number of samples per symbol were systematically varied.

5.4.1 Effect of the Parameter β

The first experiment investigated the influence of the parameter β on the bit error rate. It is important to understand why β is varied. In the literature, β is often referred to as a *security parameter*, since a smaller separation between the values in the β mapping table makes interception and correct decoding of the transmitted signal more difficult. Consequently, increasing the difference between the available β values is expected to simplify decoding while simultaneously reducing the overall security of the communication system.

The simulation was performed using the parameters *samples_per_symbol* = 1000, *L* = 20, and *noise_ratio* = 0.0. The obtained results are shown below.

Different β mapping tables were evaluated, ranging from $[-0.1, 0.1]$ to $[-1.0, 1.0]$. The quantity $\Delta\beta$ denotes the separation between the two values in each mapping table. A significant reduction in BER is observed at $\Delta\beta = 2$, where no bit errors occurred in any of the 100 simulation runs. Since a BER of zero cannot be represented on a logarithmic scale, it was replaced by the smallest measurable BER value, namely

$$\frac{1}{102400} \approx 10^{-5}.$$

The same approach was adopted for all simulations in which no bit errors were observed.

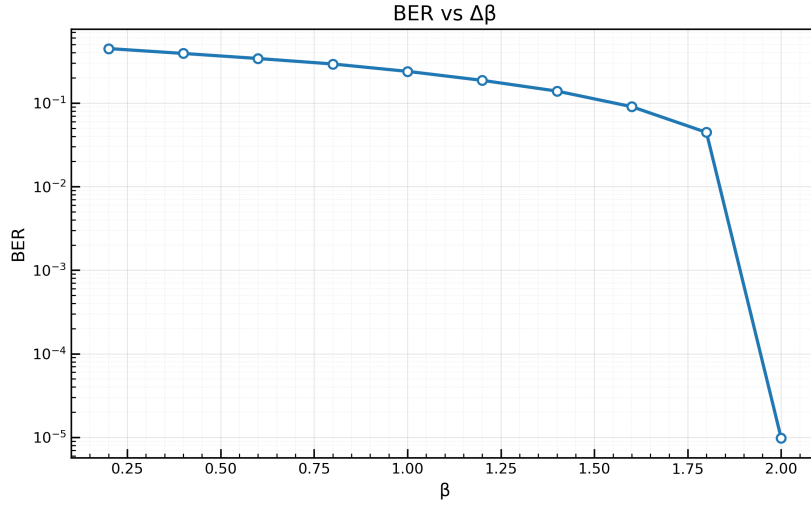


Figure 5.2: BER as a function of $\Delta\beta$

5.4.2 Effect of the Parameter γ

The second experiment investigated the influence of the parameter γ on the bit error rate. The parameter γ determines the signal-to-noise power ratio (MSNR) and therefore directly affects the quality of the received signal. Higher values of γ reduce the relative influence of noise and are therefore expected to produce lower BER values.

The simulation was performed using *samples_per_symbol* = 1000, *L* = 20, and *noise_ratio* = 1.0.

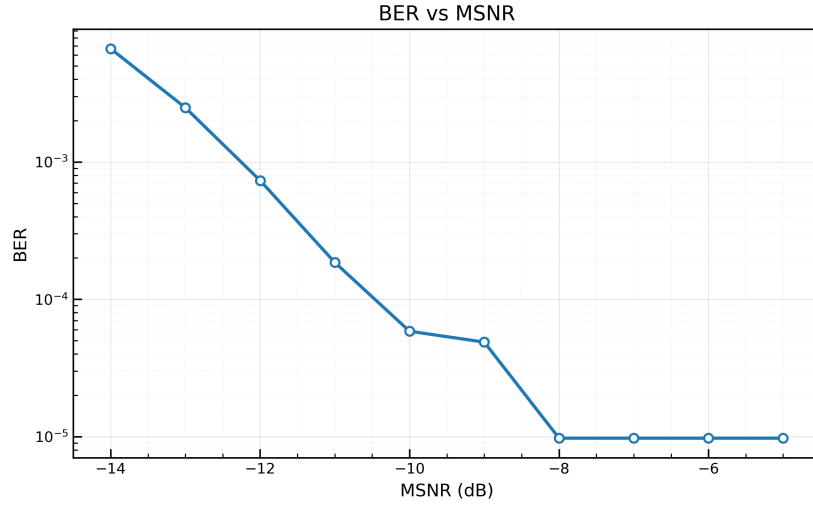


Figure 5.3: BER as a function of the parameter γ

As expected, the BER decreases as MSNR increases. Interestingly, even for the lowest investigated MSNR values, the BER remains below 10^{-2} , demonstrating that the proposed communication system maintains satisfactory reliability even under relatively unfavorable channel conditions.

The MSNR values shown in this experiment are expressed in decibels according to

$$\text{MSNR}_{\text{dB}} = 10 \log_{10} \left(\frac{\gamma}{\gamma_G} \right), \quad (5.12)$$

where γ_G denotes the γ parameter of the Gaussian distribution and is equal to 1.

Additional Gaussian noise was introduced using the built-in GNU Radio Companion channel block. The parameter `noise_ratio` was set to 1.0 because, in the absence of noise (`noise_ratio=0`), the behavior differs significantly, as shown in Figure 5.4.

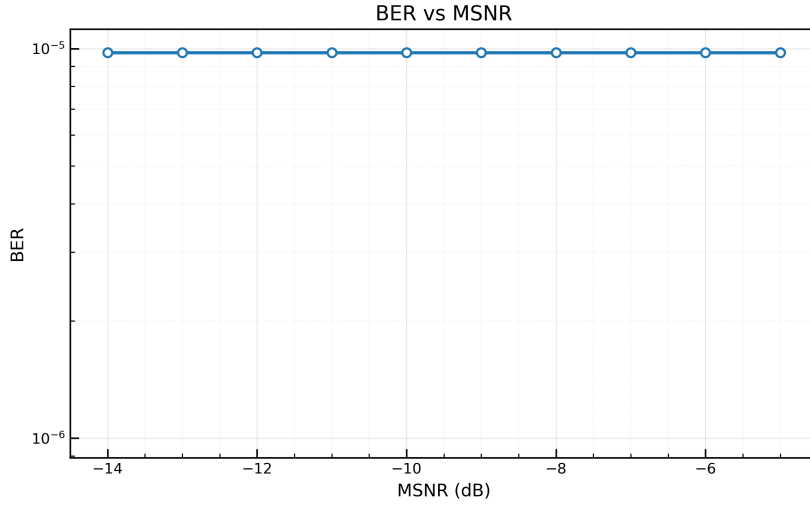


Figure 5.4: BER as a function of the parameter γ in a noise-free environment

The results obtained under ideal conditions confirm that, in the absence of noise, the parameter γ has no influence on the bit error rate, since transmission is error-free.

5.4.3 Effect of the Sample Size

In all previous experiments, the number of samples per symbol was fixed at 1000. In the final experiment, we investigated how the BER depends on the number of samples used to represent each transmitted symbol. It is expected that increasing the number of samples improves the accuracy of parameter estimation and therefore reduces the BER.

The simulation was performed using $\text{beta_map} = [-1.0, 1.0]$, $L = 8$, and $\text{noise_ratio} = 0.0$.

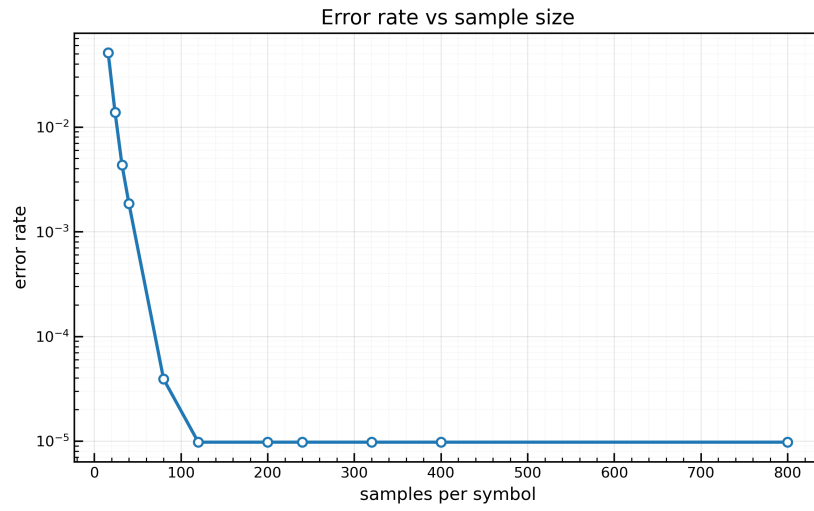


Figure 5.5: BER as a function of the number of samples per symbol

As expected, the BER is relatively high when only a small number of samples is used. However, error-free transmission is already achieved with approximately 120 samples per symbol.

Since this experiment was conducted without channel noise, it was repeated with $noise_ratio = 6.5$ to evaluate system performance under more realistic operating conditions.

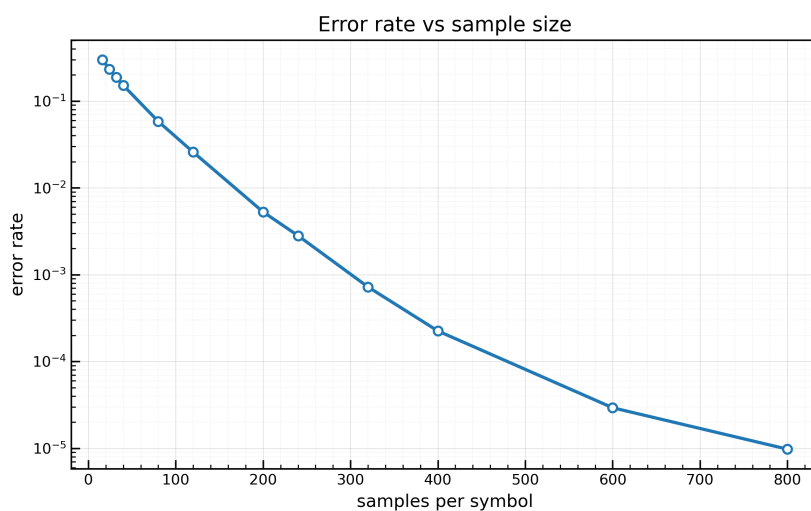


Figure 5.6: BER as a function of the number of samples per symbol in the presence of noise

The results show that, in the presence of noise, a significantly larger number of samples is required to achieve a low BER. Error-free transmission is only obtained at approximately 800 samples per symbol, which is expected since stronger noise requires more observations for reliable parameter estimation and symbol detection.

Chapter 6

Implementation and Simulation in a Real Environment

